

Ariel Lubonja

PhD Candidate, Scalable & Distributed Machine Learning. Graduating 05/28

(443) 854-2791 | ariel@cs.jhu.edu | github.com/ariellubonja | linkedin.com/in/ariellubonja | Baltimore, MD

EDUCATION

Johns Hopkins University

PhD, Computer Science — GPA: 3.92/4.0

Aug 2022 – May 2028 (Exp.)

- Advisors: Randal Burns, Joshua Vogelstein, Alan Yuille

MS, Data Science — GPA: 3.8/4.0 — Dean's List

Jan 2021 – May 2022

RESEARCH EXPERIENCE

Yggdrasil Decision Forests (Google) — Open-Source Research Collaboration May 2025 – Present

Joint work with the Google YDF team (Zurich)

- Reduced Google YDF's sparse oblique training time by 6x (12 h to 2 h) via 7 upstream pull requests.
- Designed two novel methods contributing 1.7–2.5x of this baseline: SIMD-vectorized histogram binning (two vector compares replace per-sample binary search) and an adaptive splitter that picks numerical split type by cardinality.
- Extending this work to multi-node distributed training and SIMD-vectorized inference via QuickScorer (ongoing).
- Mentored 2 graduate students (one MS, one junior PhD).

NVIDIA PhysicsNeMo Contributor

Nov 2025 – Feb 2026

2 merged pull requests

- Implemented and tested a new API in NVIDIA PhysicsNeMo; merged upstream in 2 pull requests.
- Improved runtime by 2.2x on low-resolution inputs by eliminating a redundant API call.
- Funded by and contributing to the Johns Hopkins Turbulence Database (JHTDB), the largest public turbulence-simulation archive and a PhysicsNeMo data source.

Scalable Node Embedding for Graph ML

Dec 2023 – May 2024

Open-source code and IPDPS-W 2024 paper

- Cut node embedding of the 1.8B-edge Friendster graph from 56 min to 6.4 s on 24 cores (500x vs. Python original, 17x vs. a Numba JIT baseline) via a lock-free, edge-parallel Ligra reformulation.
- Published as an open-source package and at IPDPS-W 2024 (GrAPL)

GPU-Accelerated Orthogonal Matching Pursuit

May 2022 – Jan 2023

Python package (PyPI) and arXiv preprint

- Built a drop-in GPU replacement for scikit-learn's Orthogonal Matching Pursuit (OMP), up to 310x faster.
- Applied Gramian-matrix precomputation and Cholesky factorization for up to 83x faster single-core execution.
- Implemented batching for CPU and GPU acceleration of large matrices using cuBLAS.
- Released as PyPI package batched-omp, with an arXiv preprint and open-source code.

WORK EXPERIENCE

Software Developer

May 2017 – Jan 2021

jBoxers

Sofia, Bulgaria

- Built an Alice in Wonderland chatbot for USC using Mycroft speech-to-text, an NLP model, and Dialogflow.
- Developed frontend (Angular) and backend (Django, Java EE) for the Library of Congress's Virtual Card Catalog.
- Supervised 2 summer interns.

TECHNICAL SKILLS

Languages & Tools: C++, Python, Java, CUDA, Git, Bazel

HPC & Performance: SIMD (AVX-512, Google Highway), Intel VTune, Nsight Systems, MPI, Slurm, cuBLAS, Zarr

Machine Learning & Mathematics: Tree Ensembles, Graph Learning, Sparse ML, PyTorch, Numerical Optimization

TEACHING, HONORS & LEADERSHIP

Teaching Assistant: Big Data Systems (EN.601.612), Fall 2026 — Spark, Kafka, Hadoop, NoSQL

Parallel Programming & Performance Engineering (EN.601.620), 2022–2024 — SIMD, multicore, memory, CUDA

Fellowship to Johns Hopkins University: Fulbright-style fully funded Albanian fellowship. 1 of 14 nationwide

Vice President: Upsilon Pi Epsilon (international CS honor society), Johns Hopkins chapter